N83 K1120: the padd rank stage lowers cold time by 12.49 s

Porting the rank-stage change of `aburan28/cryptanalysis` PR #970 to crypto's K1120/root11 producer (`KIC_RANK_SOLVE=padd`) lowers the mean cold one-target process time from **119.899 s to 107.412 s**, a paired saving of **12.487 s, 95% bootstrap interval [12.248, 12.723] s** (ratio 1.1163 [1.1139, 1.1186]), over 20 points on one GitHub `ubuntu-24.04` runner. The rank stage speeds up **1.2194x [1.2155, 1.2233]**, above the pre-registered 1.2 kill threshold by 0.019, so the idea transfers, with a much smaller rank gain than #970's 1.377x. Relations, rank, scalars and the factor base are identical to the current producer on all 20 points. Matched strong rho is still faster: cold IC/rho is **52.8 [45.0, 63.2]** with padd and **58.9 [50.3, 70.3]** without (60 rho draws), IC wall over rho wall, IC on 4 threads and rho on 1.

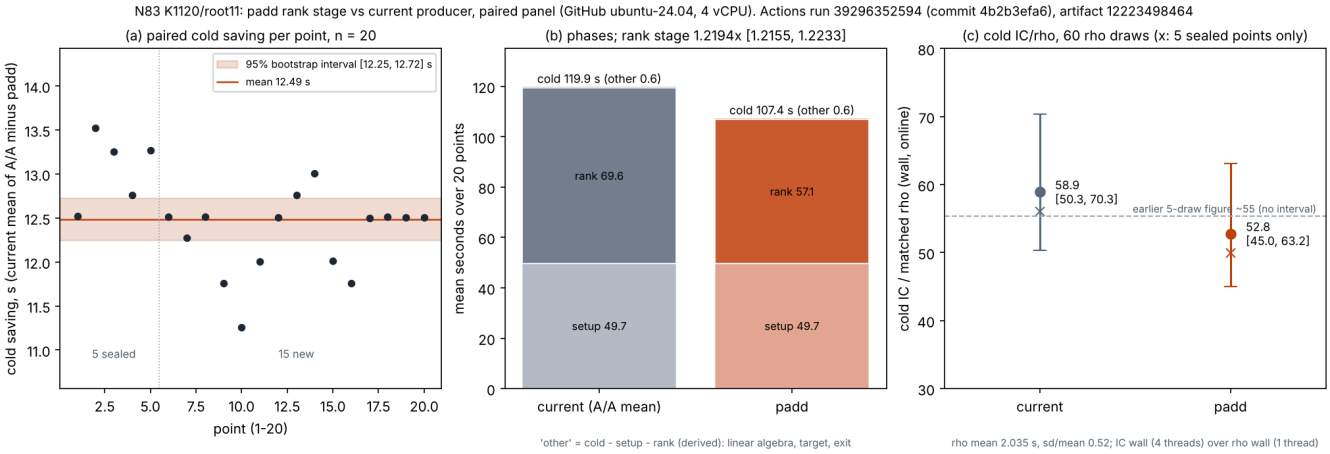


Figure 1: Paired panel. Source: plot.py, data: points.json, analysis.json. Vector: result.svg.

Result (measured; frozen protocol in PROTOCOL.md)

Quantity (mean over 20 points)	current (A/A mean)	padd	paired saving, s, 95% bootstrap	ratio current/padd, 95%
cold child wall	119.899	107.412	12.487 [12.248, 12.723]	1.1163 [1.1139, 1.1186]
rank stage (in process)	69.621	57.096	12.525 [12.330, 12.728]	1.2194 [1.2155, 1.2233]
setup (index build etc.)	49.685	49.688	−0.003 [−0.091, 0.081]	0.9999 [0.9982, 1.0017]
one-target online, ms	176.9	177.2	−0.2 [−1.4, 0.7]	0.9988 [0.9919, 1.0037]

Student-t interval (df 19) for the cold saving: [12.229, 12.745] s. All 20 per-point cold savings are positive, 11.26 to 13.52 s. The saving is the rank stage's: the rank-stage saving (12.53 s) and the cold saving (12.49 s) agree, and setup and the target scan (which keeps the S3 solve) do not move.

Controls (all passed). A/A (`cur_a` over `cur_b`): cold 0.9983 [0.9954, 1.0012], rank 0.9968 [0.9933, 0.9999], inside the frozen [0.97, 1.03]. The rank A/A upper bound sits just under 1, a 0.3% slot effect that is 40 times smaller than the effect measured. Gates: scalar equals the sealed answer for every IC arm and rho draw; per point the target relation, probes, rank attempts, relations, failures, rows without gain, rank, mean rank probes and the SHA-256 of the dumped factor base are identical across the three IC arms; every padd arm records `rank_solve = padd`, zero fallback states, zero irregular inputs and one resolved hit per rank relation. `cargo test` of the example (24 tests, including `padd_rank_tests`) passed on the runner. Local bit-identity of the new binary against a binary built from `main`, sealed point 1 at K1120, also passed (identity_check.py: 1120 resolved hits, 551 swapped pairs, 0 fallbacks); that check covers one point.

Kill condition. The rank-stage speedup point estimate is 1.2194, not below 1.2, so the condition did not fire. The margin is small; a different host or K could fall under it.

Cold IC / matched rho (measured)

Rho: `n83_rho_optimization_eval`, config `fold/64/DP6`, 3 draws per point, 60 draws, mean online **2.0354 s**, sd 1.0548 s (sd/mean 0.52, as the earlier 5-draw figure suggested). Interval: resample points and, inside each draw, rho draws.

IC arm	cold IC / rho online, 95%	vs rho child wall, 95%	5 sealed points only
current	58.9 [50.3, 70.3]	47.5 [41.7, 54.7]	56.0
padd	52.8 [45.0, 63.2]	42.5 [37.3, 49.2]	49.9

The earlier figure (cold IC/rho about 55, five rho draws, no interval, `research/n83_e2e_holdout_20261016`) is inside the current arm's interval. The distance to rho remaining is 52.8x in wall time for the best IC arm. Accounting caveat: the IC process runs 4 rank threads, the rho process runs one thread (no threading in its source), so the ratio is wall over wall, not per core. Per-core accounting was not measured.

Section 5a (pricing)

IC native work per run (mean over 20 points, identical in both IC arms): 566,791,207 rank probes plus 560,285 target probes, each probe one canonicalisation and one table query, one root solve per two probes. Rho (mean of 60): 8,248,365 group additions, 8,248,364 canonicalisations, 8,117,869 partition hashes, 128,894 table queries, 128,893 inserts.

weight, GAE per native op	rho cost, GAE	IC unpriced native work, GAE (both arms)
0	8.25e6	0
0.1	9.91e6	1.42e8
1	2.49e7	1.42e9

The IC producer does not emit its group-addition count (the lifts and target descent), so its total cost in GAE cannot be formed: a named counter without an emitted count cannot be ranked, and this is recorded as an implementation gap. IC and rho are therefore **unresolved at the 0 weight boundary** and no IC-versus-rho GAE ranking is claimed. The two IC arms have identical native-operation counts, so in counted S the change is a tie; the gain is an implementation (wall-clock) gain at equal counted operations. The field work inside a root solve falls (a point addition per candidate in place of the S3 solve), which the counters above do not price.

One canonical N83 baseline row

Two N83 baselines disagreed because they measure different pipelines on different runner configurations, not because either is wrong:

	crypto frontier (this study)	cryptanalysis `docs/ic/BASELINES.md` (#970)
pipeline	K1120/root11, Bloom, sparse canonical, 4 rank threads	K600 stacked producer, one pinned CPU
cold time	119.90 s current, **107.41 s with padd**	261.0 s, 195.7 s with rank cut
rank-stage gain	1.2194x	1.377x
matched rho	strong rho, mean 2.035 s (60 draws)	tuned rT770, about 2.98 s implied
cold IC/rho	**52.8 [45.0, 63.2]** (padd)	65.7 [55.4, 80.2]

The canonical N83 row is this study's K1120/root11 + padd row on the a = 1 curve (`EC1N83Ckb1a1haacb69ee9f37`, 53-bit subgroup, 4-vCPU runner): cold 107.412 s, rho 2.035 s, ratio 52.8 [45.0, 63.2]. Its predecessor (current producer, 119.899 s, 58.9 [50.3, 70.3]) replaces the five-draw figure of about 55 with an interval. The cryptanalysis row is a different, slower configuration (K600, one CPU); the two intervals overlap but the rows are not interchangeable and neither is a rho-matched comparison of the other. `aburan28/cryptanalysis` could not be edited from this session (read-only); its `BASELINES.md` should cite this row, with the pipeline and thread difference above.

Limits

- **No extrapolation to ECC2K-130.** Every complete N83 result, this one included, is on the a = 1 curve with a 53-bit prime subgroup. The a = 0 curve (81-bit subgroup) is the ECC2K-130 analogue and has no complete IC result. Nothing here is evidence about it or about an exponent.
- One runner allocation. The intervals are over points; between-runner variance was not sampled. Earlier panels of the same pipeline on other runner allocations gave cold 119.83 s (5 points) and 119.70 s (16 points), consistent with this panel's 119.90 s.
- The child wall time is quantized to 0.5 s by the runner's reaping loop (fractional parts cluster at .1 and .6), so per-point cold savings are coarse; the in-process rank timer is not quantized and agrees.
- Fifteen new points are disjoint from every N83 point file in the repository and the 1000 derived wide3 points; earlier fixtures held only in an external archive could not be replayed (collision probability about 1e-12, not checked).
- Wall-clock is a practicality measurement, not the headline metric (AGENTS.md section 6).
- Hypothesis, not tested: #970's baseline loop pipelines table lookups with prefetch, so root arithmetic was a larger share of its per-probe cost (its profile puts the S3 solve at about 110-130 of 256 ns per probe); crypto's loop looks each key up as it is computed, so the same arithmetic saving buys less. A pipelined crypto loop is a separate candidate.

Evidence

- Run: Actions 39296352594, commit `4b2b3efa6210ce3724f1350c3a9b1cc2df1309fd` (the later commit `ceeb652a1` only reformats the added code). Artifact `12223498464`, ZIP SHA-256 `41e1a7f4c7ee33c71902b39d21c52189abb0b5f2e7f9a15fbdf910b082bd9f19` (866 MB unpacked, 265 MB zip; not committed).
- Committed here: analysis.json (also `analysis.log`), points.json, `host.json`, `source.json`, `run.log`, PROTOCOL.md, run.py, analyze.py, generate.py, extract_points.py, identity_check.py.
- Source: `examples/koblitz_orbit_dlp_fast.rs`, `scan_state128_padd`, `probe_candidate128`, `extract128_with`, tests `padd_rank_tests`.
- Host: four AMD EPYC vCPUs, 16,377,684 KiB, peak IC RSS 1,955.6 MiB; receipts in `host.json`.